IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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GitHub repository: https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Collected 90 Falcon 9 launch records via the SpaceX REST API and Wikipedia web scraping, then wrangled them into a clean, labeled dataset
- Explored the data with visualizations and SQL to uncover how payload mass, launch site, orbit type, and flight number relate to landing success
- Built an interactive Folium map and a Plotly Dash dashboard to visually analyze launch sites and outcomes
- Tuned four classification models with GridSearchCV — the Decision Tree classifier performed best, reaching 94.4% accuracy on the test set
- Key finding: launch success climbed steadily after 2013, and KSC LC-39A has the highest per-site success ratio (82%)

Introduction

- Project background and context
  - SpaceX advertises Falcon 9 launches at $62M, versus $165M+ from competitors — largely because SpaceX reuses the first stage
  - If we can predict whether the first stage will land, we can estimate the cost of a launch
- Problems we want to find answers to
  - Which variables (payload mass, launch site, orbit, flight number, booster version) most affect landing success?
  - Can a machine learning classifier reliably predict the landing outcome before launch?
  - Which launch sites and orbit types are associated with the highest success rates?

Part 1

Methodology

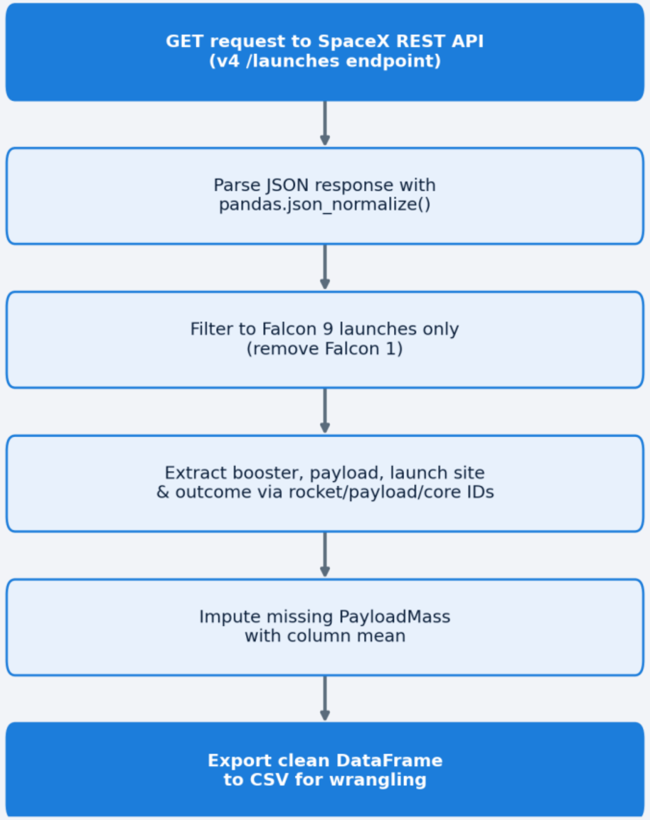
Data collection: SpaceX REST API + Wikipedia web

Data Collection

- Two complementary data sources were combined for robustness: the official SpaceX REST API and a Wikipedia table of Falcon 9 launches
- Both sources were parsed into pandas DataFrames, restricted to Falcon 9 (excluding Falcon 1), and cross-checked against each other
- Key phrases: GET request → JSON normalize → filter Falcon 9 → decode booster/payload/core IDs → BeautifulSoup HTML parse → table extraction

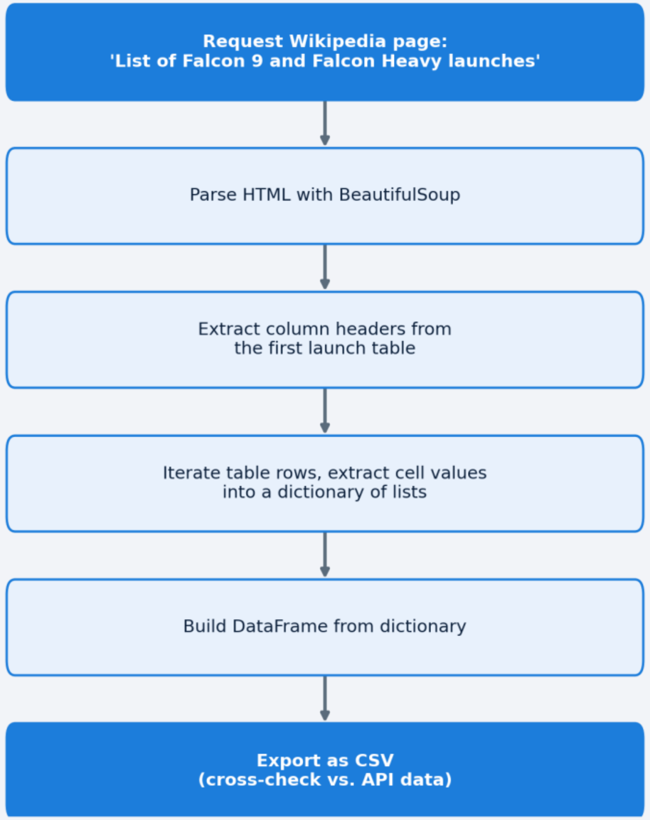
Data Collection – SpaceX API

- Sent a GET request to the SpaceX v4 REST API /launches endpoint
- Used pandas.json_normalize() to flatten nested JSON (rocket, payloads, launchpad, cores)
- Filtered to Falcon 9 only and decoded booster/payload/launchpad IDs into readable columns
- Replaced missing PayloadMass values with the column mean
- GitHub notebook: https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb



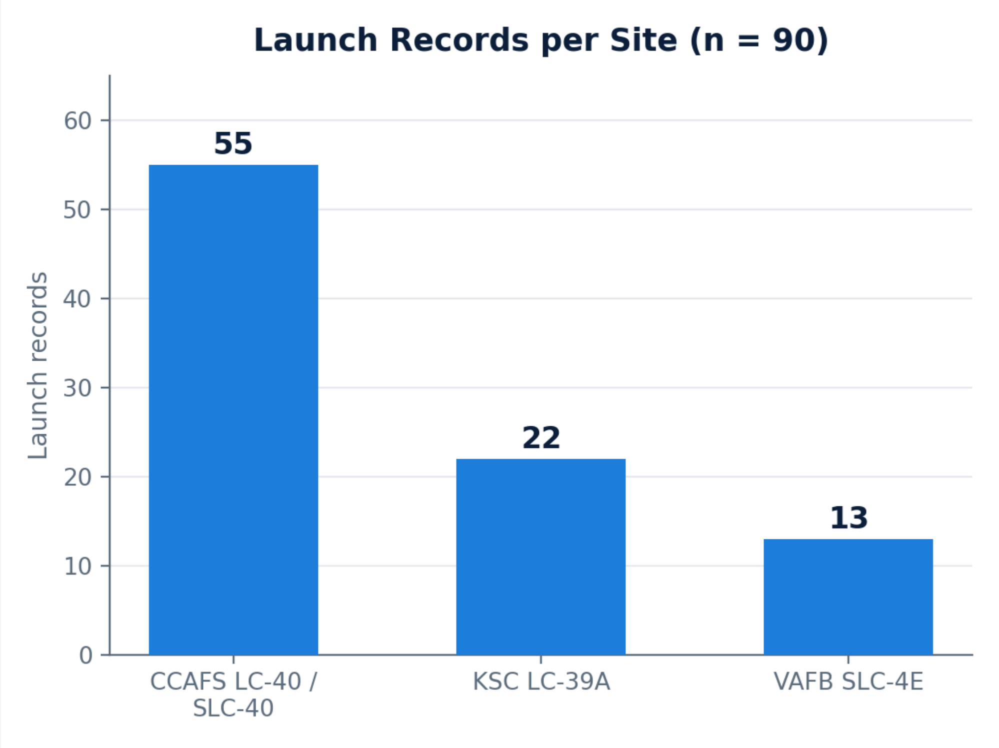
Data Collection - Scraping

- Requested the Wikipedia page "List of Falcon 9 and Falcon Heavy launches"
- Parsed the HTML with BeautifulSoup and extracted the launch table's column headers
- Iterated table rows to build a dictionary of lists, then converted it into a DataFrame
- Exported the scraped table as CSV to cross-check against the API dataset
- GitHub notebook: https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project/blob/main/jupyter-labs-webscraping.ipynb



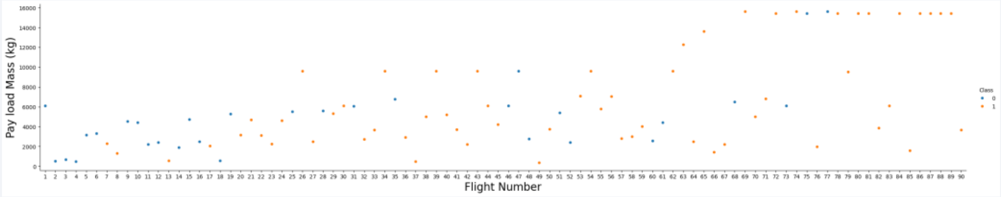
Data Wrangling

- Loaded the combined dataset: 90 Falcon 9 launches, 17 features
- Created the binary Class outcome label (1 = successful landing, 0 = unsuccessful) from the Outcome column
- Site distribution: CCAFS LC-40/SLC-40 (55), KSC LC-39A (22), VAFB SLC-4E (13)
- Applied one-hot encoding to Orbit, LaunchSite, LandingPad, and Serial
- Exported the wrangled dataset for EDA and modeling
- GitHub: https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb



EDA with Data Visualization

- Used Seaborn catplot/barplot/line charts to explore relationships between FlightNumber, PayloadMass, LaunchSite, Orbit, and landing Class
- Charted: Flight Number vs Launch Site, Payload vs Launch Site, Success Rate vs Orbit, Flight Number vs Orbit, Payload vs Orbit, and Yearly Success Trend
- These charts guided feature selection for the predictive models
- GitHub: https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project/blob/main/edadataviz.ipynb



EDA with SQL

- Loaded the SpaceX dataset into a local SQLite database (my_data1.db) via pandas.to_sql
- Used the %sql magic (ipython-sql) to run 10 exploratory queries: filtering, aggregation (SUM / AVG / MAX), date filtering, subqueries, and ranking with GROUP BY / ORDER BY
- Answered questions on total/average payload mass, landing outcomes, and booster versions
- GitHub: https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Added a folium.Map centered on the launch sites, with a Marker and labeled Circle for each site
- Colored each launch record's marker green (success) or red (failure), grouped with MarkerCluster, based on the Class label
- Calculated distances from KSC LC-39A to the coastline, nearest highway, railway, and city with the haversine formula, and drew PolyLine connectors
- GitHub: https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Built a Dash app with a dropdown to filter by launch site and a range slider for payload mass
- Added a pie chart of total successful launches, switchable between all sites and a single selected site
- Added a scatter chart of Payload Mass vs. launch outcome, colored by Booster Version Category, that updates live with the range slider
- Dashboard developed and run locally per the lab instructions

Predictive Analysis (Classification)

- Standardized features with StandardScaler and split the data 80/20 (random_state = 2)
- Tuned four classifiers with GridSearchCV (cv=10): Logistic Regression, SVM, Decision Tree, KNN
- Evaluated each tuned model on the held-out test set using accuracy and a confusion matrix
- Selected the Decision Tree as the best model (94.4% test accuracy)
- GitHub: https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project/blob/main/SpaceX%20Machine%20Learning%20Prediction%20Part_5.ipynb

Results

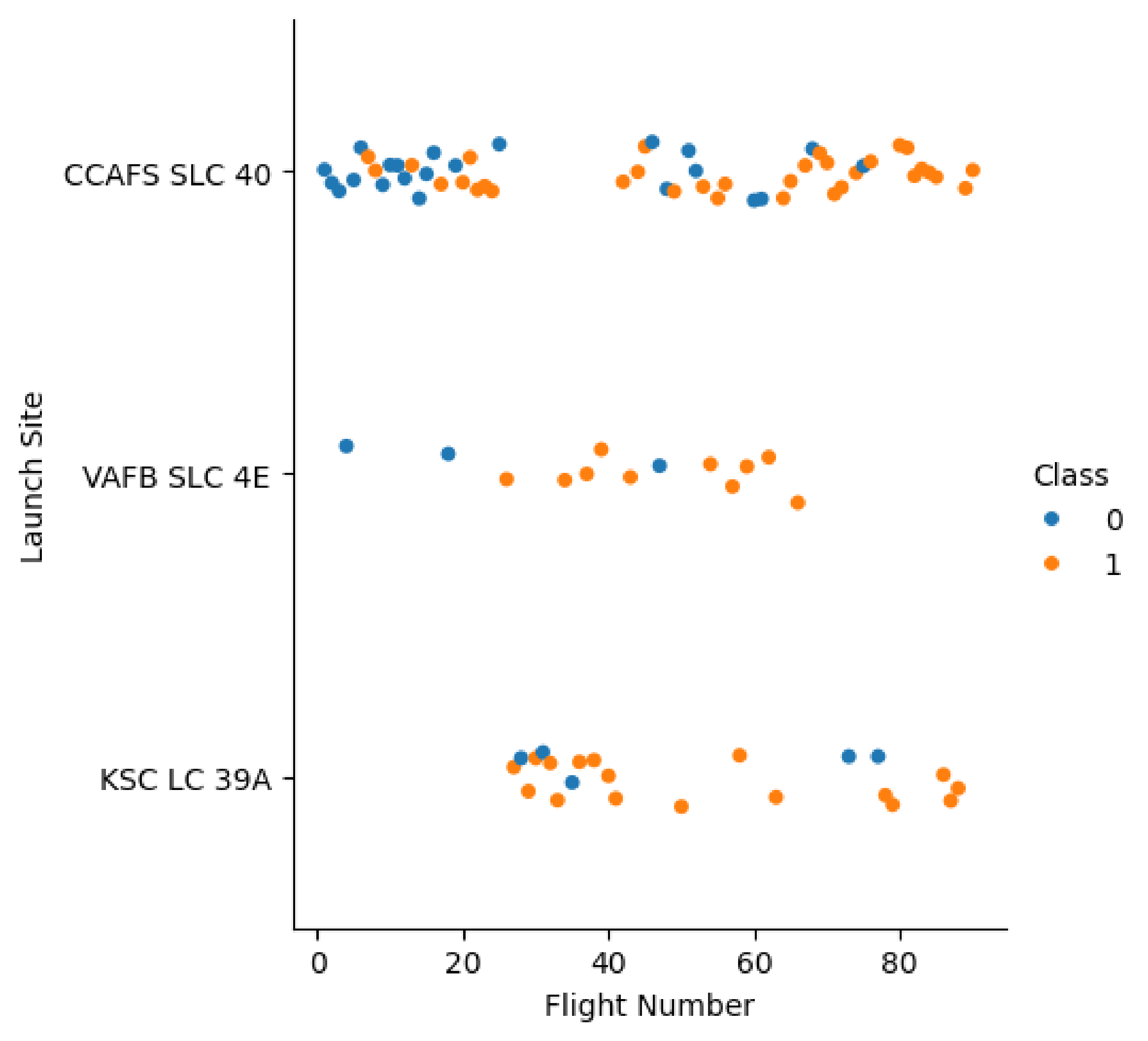
- Exploratory data analysis results (visualization + SQL)
- Interactive analytics: Folium map and Plotly Dash dashboard screenshots
- Predictive analysis results: model comparison and confusion matrix

Part 2

Insights drawn from EDA

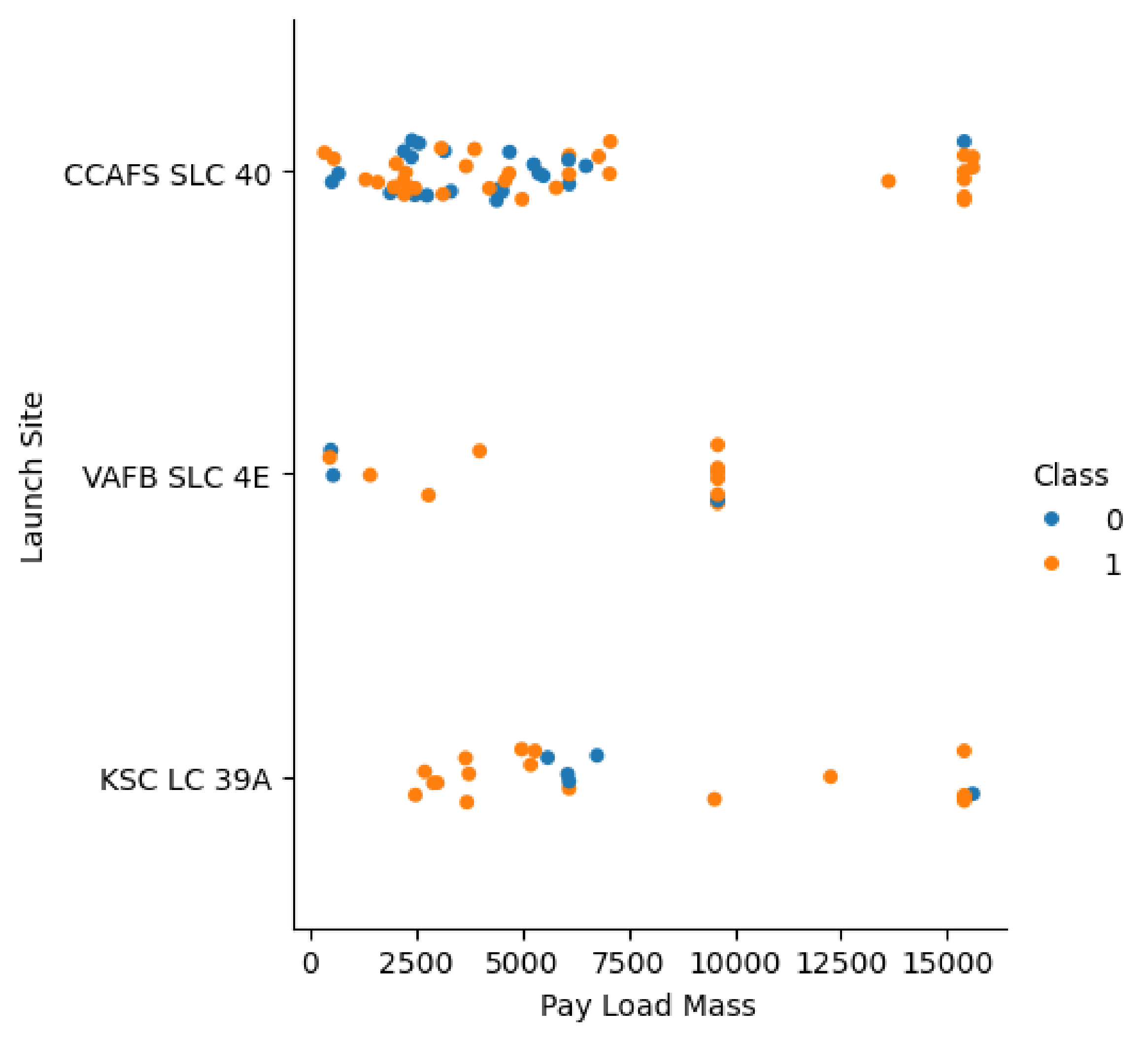
Flight Number vs. Launch Site

- Each launch site shows a mix of successful (1) and unsuccessful (0) landings across the flight sequence
- Success becomes noticeably more frequent at higher flight numbers, especially at CCAFS SLC 40 and KSC LC-39A — reflecting SpaceX's learning curve over time



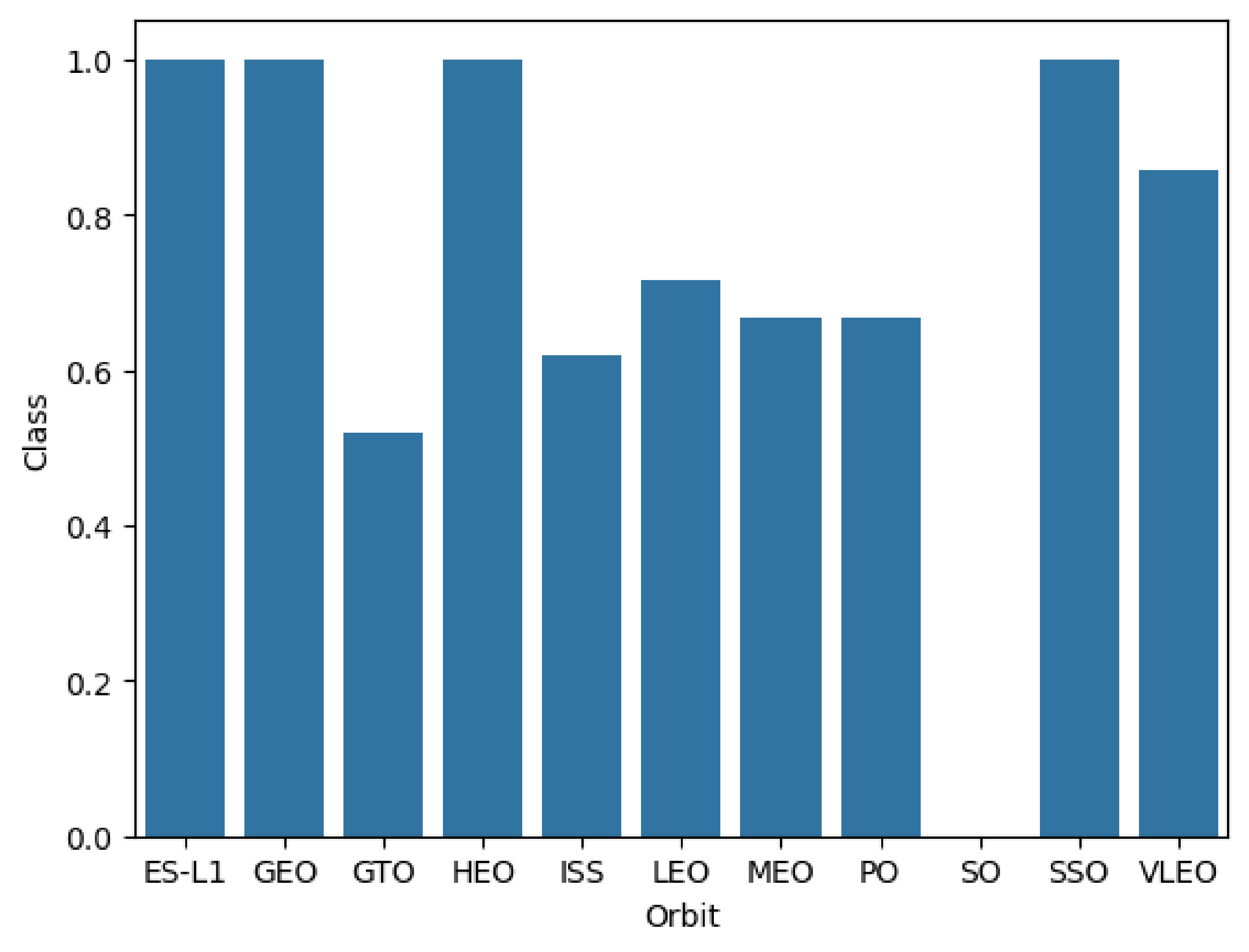
Payload vs. Launch Site

- VAFB SLC-4E has never launched a payload heavier than roughly 10,000 kg in this dataset
- CCAFS pads handle the widest spread of payload masses, including the heaviest missions



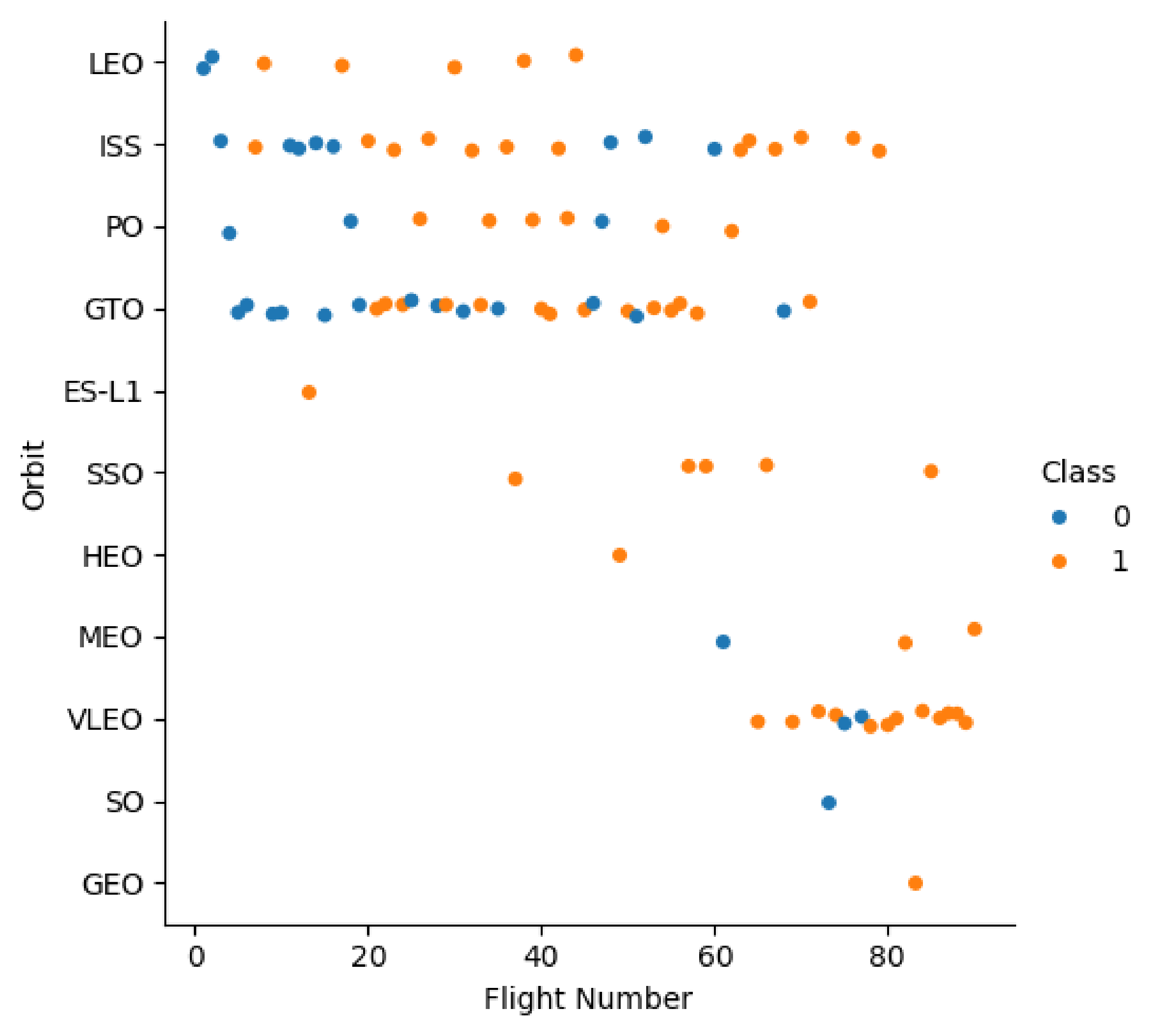
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, and SSO orbits show a 100% success rate in this dataset
- GTO shows a comparatively lower and more mixed success rate



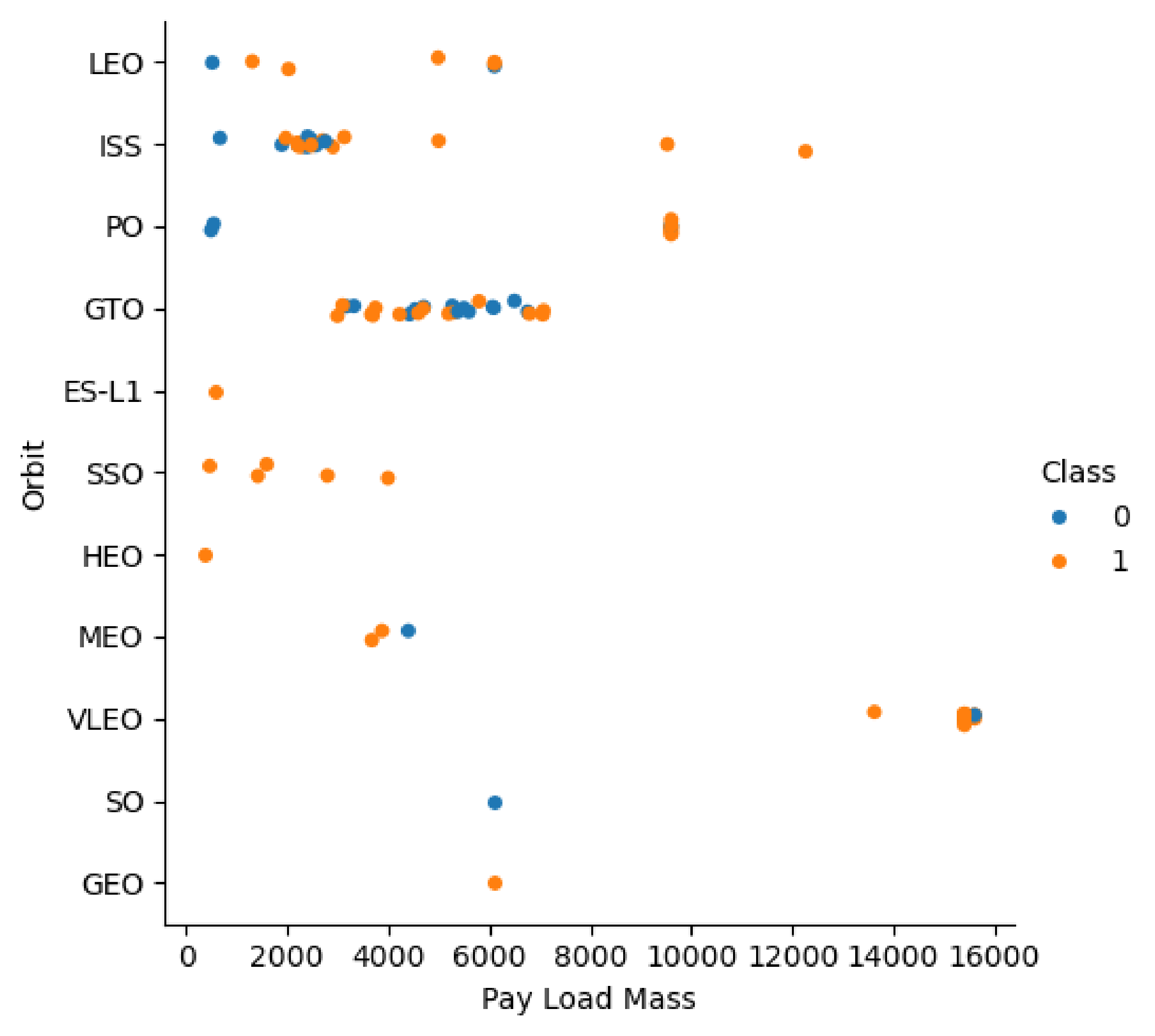
Flight Number vs. Orbit Type

- In LEO, success correlates strongly with flight number — later flights land more reliably
- In GTO, flight number shows little relationship with landing success



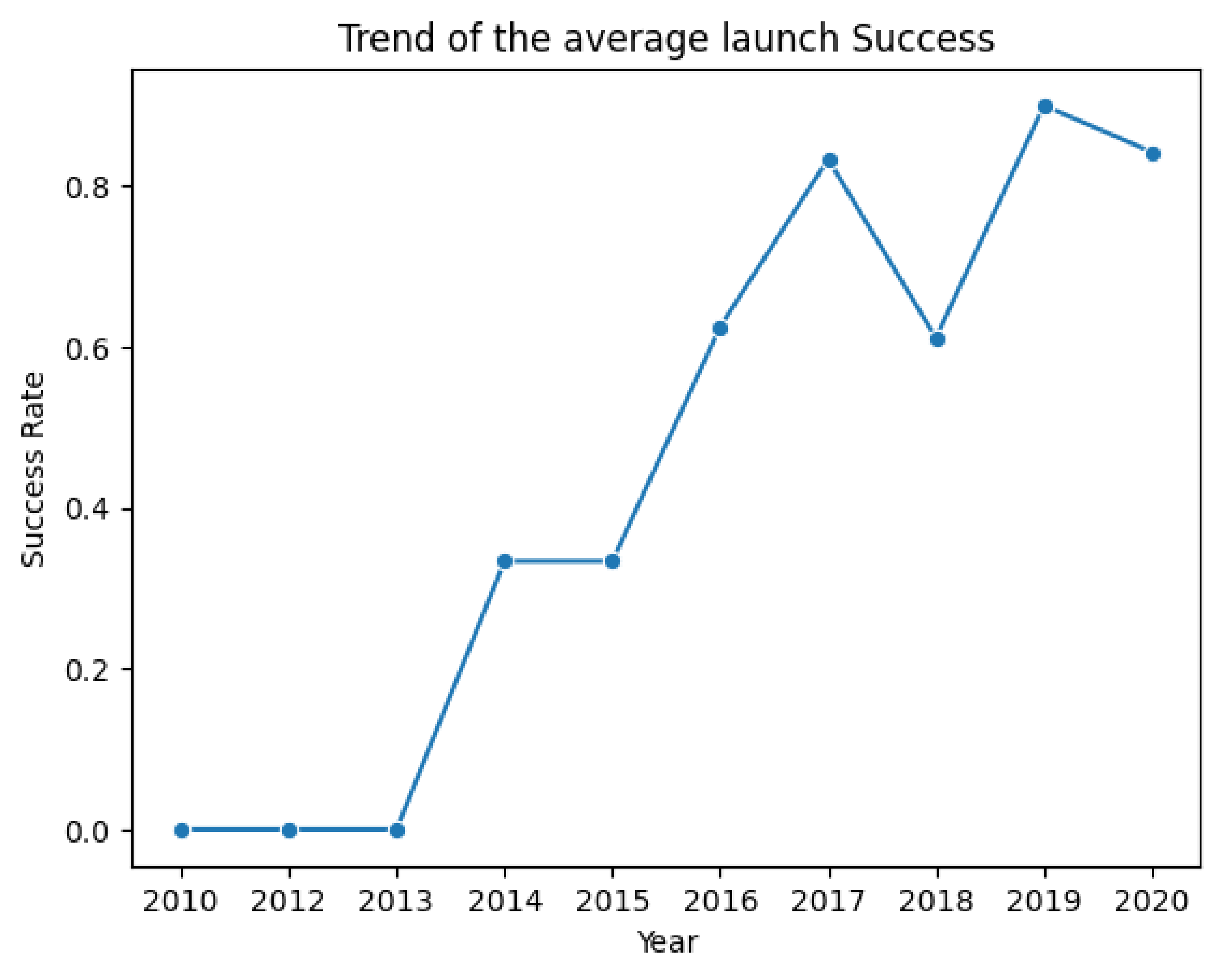
Payload vs. Orbit Type

- Heavier payloads to Polar, LEO, and ISS orbits succeed more often
- For GTO, payload mass alone doesn't clearly separate successful from unsuccessful landings



Launch Success Yearly Trend

- Success rate rose steadily from 2013 through 2020 as SpaceX refined its landing techniques
- Early years (2010–2012) show 0% landing success, since landing attempts hadn't begun yet



All Launch Site Names

- Query: SELECT DISTINCT Launch_Site FROM SPACEXTBL
- Result: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
- Four distinct launch pads were used across the mission history in this dataset

Launch Site Names Begin with 'CCA'

- Query: SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
- Result: the first 5 CCAFS launches, including the 2010-06-04 Dragon Qualification Unit flight and the 2010-12-08 COTS demo flight
- Confirms wildcard filtering correctly isolates all Cape Canaveral pads (LC-40 and SLC-40)

Total Payload Mass

- Query: SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)'
- Result: 45,596 kg total payload mass carried on NASA Commercial Resupply Services missions

Average Payload Mass by F9 v1.1

- Query: SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1'
- Result: 2,928.4 kg average payload mass for the F9 v1.1 booster version

First Successful Ground Landing Date

- Query: SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success'
- Result: 2018-07-22 — the first successful ground-pad landing recorded in the dataset

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query: filter Landing_Outcome = 'Success(drone ship)' AND payload strictly between 4000–6000 kg
- Result: no rows matched — the actual outcome string in this table is 'Success (drone ship)' with a space
- Insight: a good reminder to always verify exact category strings before filtering categorical text columns in SQL

Total Number of Successful and Failure Mission Outcomes

- Query: SELECT COUNT(Mission_Outcome) FROM SPACEXTBL
- Result: 101 total mission-outcome records loaded into the SPACEXTBL table

Boosters Carried Maximum Payload

- Query: subquery WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)
- Result: 12 F9 B5 boosters tied for the maximum payload mass, including B1048.4, B1049.4, B1051.3, B1056.4, and B1060.3

2015 Launch Records

- Query: filter Landing_Outcome LIKE 'Failure (drone ship)' AND substr(Date,1,4) = '2015'
- Result: two drone-ship landing failures in 2015 — both F9 v1.1 boosters launching from CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Query: GROUP BY Landing_Outcome, COUNT(*) ORDER BY count DESC, for dates between 2010-06-04 and 2017-03-20
- Result (descending): No attempt (10), Success/Failure (drone ship) (5 each), Success (ground pad) (3), Controlled (ocean) (3), Uncontrolled (ocean) (2), Failure (parachute) (2), Precluded (drone ship) (1)

Part 3

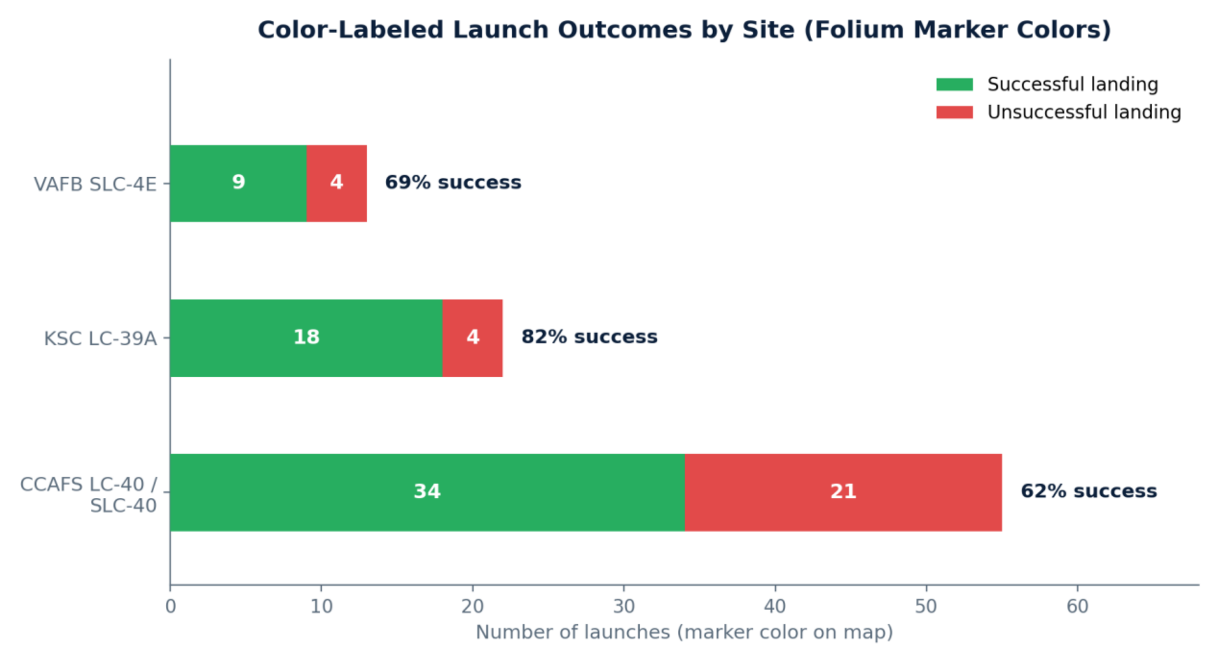
Launch Sites Proximities Analysis

All Launch Sites on the Global Map

- Added a marker and labeled circle for each of the 3 launch-site clusters (4 pads) in the dataset
- CCAFS LC-40/SLC-40 accounts for the most launches (55), followed by KSC LC-39A (22) and VAFB SLC-4E (13)

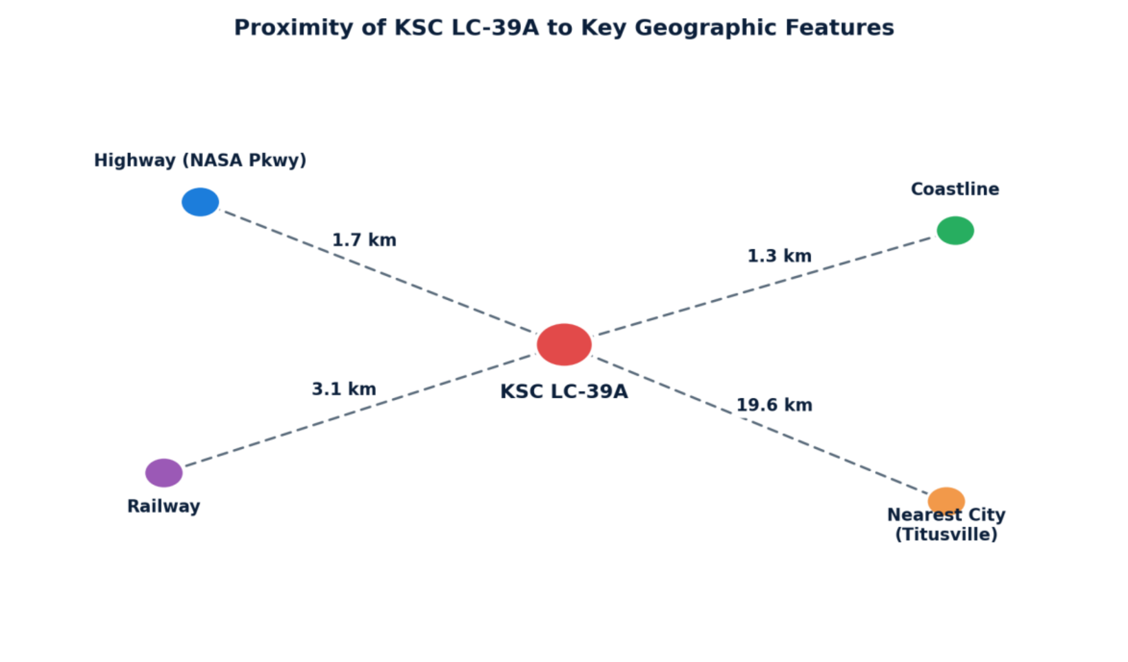
Color-Labeled Launch Outcomes

- Markers are colored green for a successful landing and red for an unsuccessful one, grouped by site
- KSC LC-39A shows the highest success ratio (82%), while CCAFS LC-40/SLC-40 — used for the earliest, least-refined missions — shows the most failures



KSC LC-39A Proximity Analysis

- Measured straight-line distance from the launch pad to the coastline, nearest highway, railway, and city using the haversine formula
- The pad sits close to the coastline (~1.3 km) for safety during ascent, yet still within a few km of the infrastructure launch operations depend on

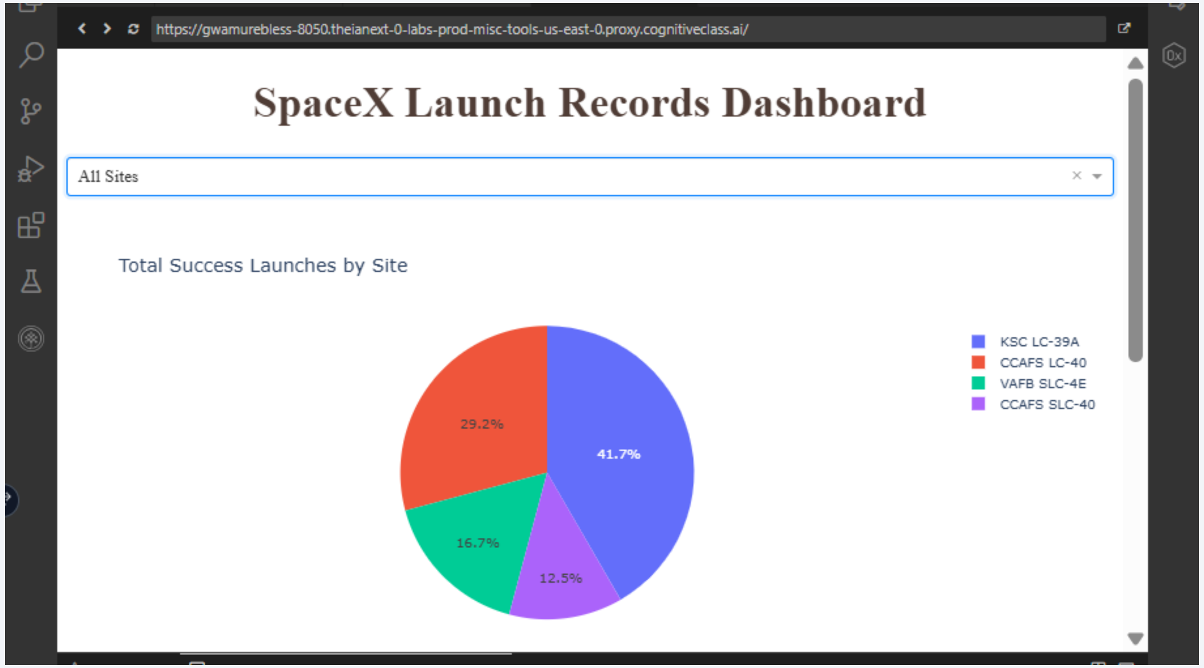


Part 4

Build a Dashboard with Plotly Dash

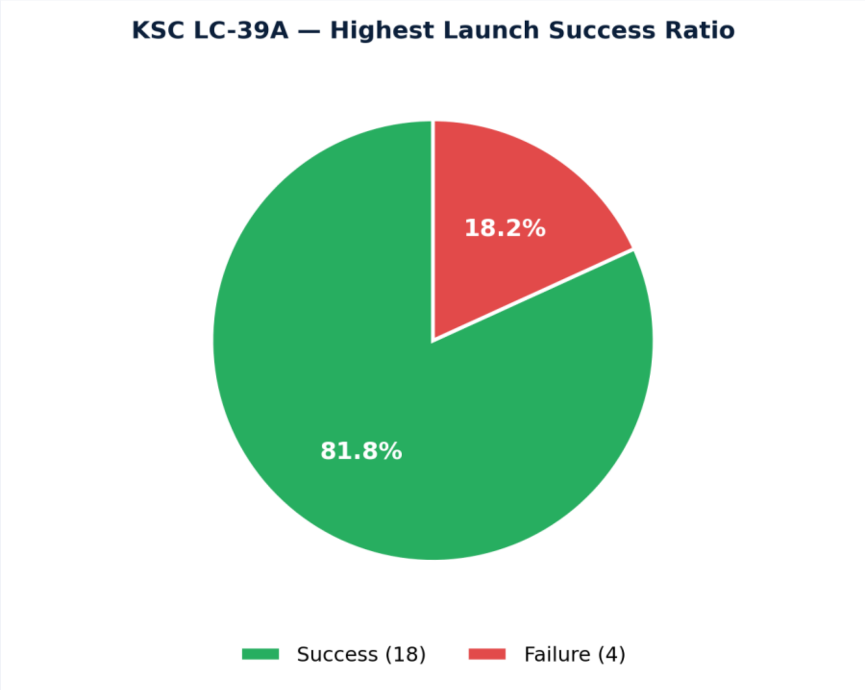
Launch Success Count by Site

- Pie chart shows the share of total successful launches contributed by each site
- KSC LC-39A contributes 41.7% of all successful launches despite hosting fewer total missions than CCAFS — reflecting its higher success rate



Highest Success-Ratio Site: KSC LC-39A

- Filtering the dashboard to KSC LC-39A alone shows an 81.8% success rate for that site
- This is the highest per-site success ratio of the three launch locations



Payload Range vs. Launch Outcome

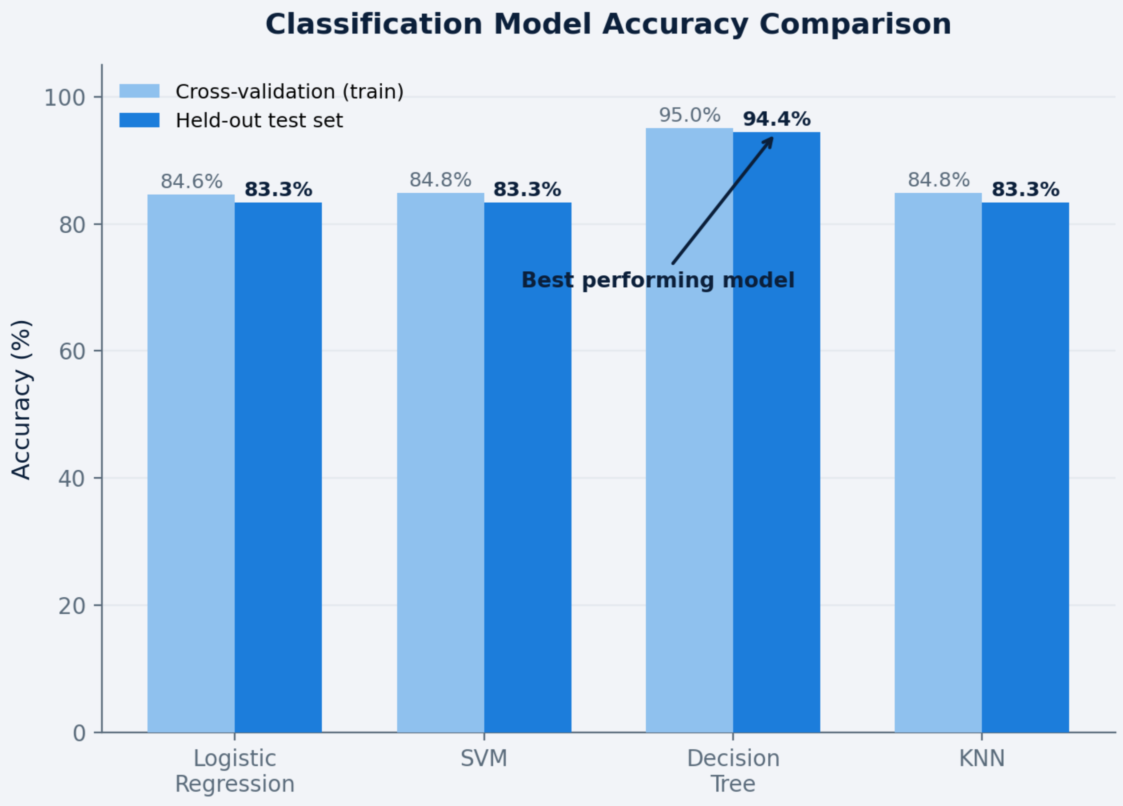
- The range slider filters launches by payload mass (0–10,000 kg) and plots outcome (0/1) by Booster Version Category
- Low-to-mid payloads (0–4,000 kg) on FT and B4/B5 boosters show the strongest concentration of successful landings

Part 5

Predictive Analysis (Classification)

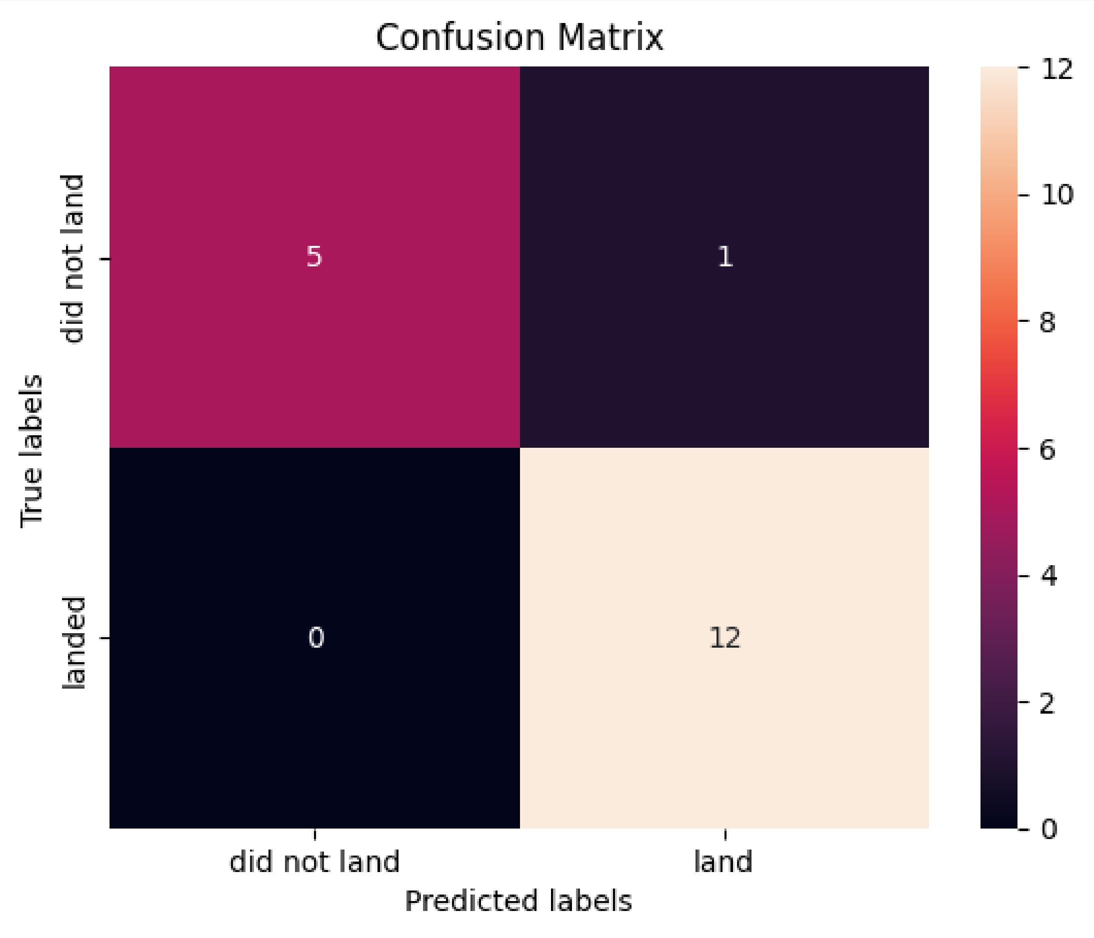
Classification Accuracy

- Logistic Regression: 84.6% CV / 83.3% test accuracy
- SVM: 84.8% CV / 83.3% test accuracy
- Decision Tree: 95.0% CV / 94.4% test accuracy — best performer
- KNN: 84.8% CV / 83.3% test accuracy



Confusion Matrix

- The Decision Tree model correctly classified 17 of 18 test launches
- Only 1 misclassification: a launch that did not land was predicted as landed (false positive); zero missed landings (false negatives)
- This strong recall on the minority "did not land" class makes it a reliable choice for pre-launch cost estimation



Conclusions

- Data collection via API + web scraping, combined with careful wrangling, produced a clean, analysis-ready dataset of 90 Falcon 9 launches
- Payload mass, orbit type, launch site, and flight number all show a visible relationship with landing success in the EDA
- KSC LC-39A has both the most reliable historical success rate (82%) and the largest share of total successful launches
- A tuned Decision Tree classifier predicts landing outcome with 94.4% test accuracy, the best of four models evaluated
- These results support SpaceX's ability to reliably reuse first stages — directly informing cost estimates for competing launch bids

Appendix

- Full code, SQL queries, and notebook outputs are available in the GitHub repository linked on the title slide
- https://github.com/ThanyaniMunothaJr/IBM-SpaceX-Capstone-Project
- Includes: SpaceX API collection & web scraping notebooks, data wrangling notebook, EDA visualization & SQL notebooks, Folium map notebook, and the machine learning prediction notebook


Thank you!